

Database Schema Proposal

Based on PRD and discussion, we have designed a high level of Database Schema. You can find it attached with this document.

admin_users				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
uuid	CHAR(36)	UNIQUE	No	Public-safe unique identifier
first_name	VARCHAR(100)		No	Admin first name
last_name	VARCHAR(100)		No	Admin last name
email	VARCHAR(255)	UNIQUE	No	Admin login email
password_hash	VARCHAR(255)		No	Secure password hash
mobile_number	VARCHAR(20)		Yes	Optional phone number
status	ENUM('active','inactive','locked')		No	Admin account status
last_login_at	DATETIME		Yes	Last successful login
password_changed_at	DATETIME		Yes	Last password change/reset
failed_login_attempts	INT	DEFAULT 0	No	Counter for failed login attempts
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
deleted_at	DATETIME		Yes	Soft delete timestamp
id	BIGINT UNSIGNED	PK	No	Primary key
name	VARCHAR(100)	UNIQUE	No	Role name (e.g. Super Admin)
code	VARCHAR(50)	UNIQUE	No	System role code

description	TEXT		Yes	Role description
is_system_role	TINYINT(1)	DEFAULT 1	No	Marks predefined role
status	ENUM('active','inactive')		No	Role status
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
admin_permissions				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
module_name	VARCHAR(100)		No	Module name such as pricing or tools
permission_key	VARCHAR(150)	UNIQUE	No	Permission key like pricing_tier.create
description	TEXT		Yes	Permission meaning
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
admin_role_permissions				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
admin_role_id	BIGINT UNSIGNED	FK -> admin_roles.id	No	Assigned admin role
admin_permission_id	BIGINT UNSIGNED	FK -> admin_permissions.id	No	Assigned permission
created_at	DATETIME		No	Record creation timestamp

admin_user_roles				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
admin_user_id	BIGINT UNSIGNED	FK -> admin_users.id	No	Admin user
admin_role_id	BIGINT UNSIGNED	FK -> admin_roles.id	No	Role granted to admin user
created_at	DATETIME		No	Record creation timestamp
pricing_tiers				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
name	VARCHAR(100)		No	Tier name like Free, Pro, Enterprise
code	VARCHAR(50)	UNIQUE	No	Unique tier code
description	TEXT		Yes	Tier details
monthly_price	DECIMAL(12,2)		No	Monthly subscription amount
yearly_price	DECIMAL(12,2)		Yes	Optional annual price
currency_code	CHAR(3)	DEFAULT 'USD'	No	Billing currency
max_organization_seats	INT		Yes	Max users allowed in organization
max_api_keys_per_org	INT		Yes	Max org-level API keys
max_api_keys_per_user	INT		Yes	Max user-level API keys
max_monthly_events	BIGINT		Yes	Monthly telemetry cap

max_monthly_exports	INT		Yes	Monthly export limit
has_alerts_feature	TINYINT(1)	DEFAULT 1	No	Whether alerts feature is enabled
has_csv_export	TINYINT(1)	DEFAULT 1	No	Whether CSV export is enabled
has_tool_breakdown	TINYINT(1)	DEFAULT 1	No	Whether tool/model analytics are enabled
sort_order	INT	DEFAULT 0	No	UI sorting order
status	ENUM('active','inactive')		No	Tier status
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
deleted_at	DATETIME		Yes	Soft delete timestamp
pricing_tier_feature_overrides				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
pricing_tier_id	BIGINT UNSIGNED	FK -> pricing_tiers.id	No	Related pricing tier
feature_key	VARCHAR(150)		No	Feature code like alerts.email
feature_value	VARCHAR(255)		No	Stored feature value
value_type	ENUM('string','integer','decimal','boolean','json')		No	Type of feature value
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
third_party_tools				

Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
name	VARCHAR(100)		No	Display name
code	VARCHAR(50)	UNIQUE	No	Tool code like claude or gemini
vendor_name	VARCHAR(100)		No	Vendor/provider name
category	VARCHAR(100)		Yes	Category like coding assistant or CLI
logo_url	VARCHAR(500)		Yes	Logo asset URL
documentation_url	VARCHAR(500)		Yes	Official documentation URL
integration_type	ENUM('api_key','oauth','manual','hybrid')		No	Integration mode
status	ENUM('active','inactive','beta')		No	Tool support status
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
third_party_tool_configs				

Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
third_party_tool_id	BIGINT UNSIGNED	FK -> third_party_tools.id	No	Related tool
config_key	VARCHAR(150)		No	Config name like endpoint_url
config_value	TEXT		No	Config value
value_type	ENUM('string','integer','decimal','boolean','json')		No	Type of config value
is_secret	TINYINT(1)	DEFAULT 0	No	Whether value should be encrypted/hidden

status	ENUM('active','inactive')		No	Config status
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
admin_audit_logs				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
admin_user_id	BIGINT UNSIGNED	FK -> admin_users.id	Yes	Actor admin user
action_type	VARCHAR(100)		No	Action such as CREATE, UPDATE, LOGIN
module_name	VARCHAR(100)		No	Affected module
record_type	VARCHAR(100)		No	Target entity/table name
record_id	VARCHAR(100)		No	Target record id
old_values_json	JSON		Yes	Previous state snapshot
new_values_json	JSON		Yes	Updated state snapshot
ip_address	VARCHAR(45)		Yes	IPv4/IPv6 address
user_agent	TEXT		Yes	Request user agent
created_at	DATETIME		No	Log creation timestamp
users				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
uuid	CHAR(36)	UNIQUE	No	Public-safe unique identifier
first_name	VARCHAR(100)		No	User first name

last_name	VARCHAR(100)		No	User last name
email	VARCHAR(255)	UNIQUE	No	User signup/login email
password_hash	VARCHAR(255)		Yes	Password hash; nullable for future SSO/OAuth
email_verified_at	DATETIME		Yes	Email verification timestamp
status	ENUM('pending','active','inactive','suspended')		No	User account status
last_login_at	DATETIME		Yes	Last successful login
failed_login_attempts	INT	DEFAULT 0	No	Counter for failed login attempts
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
deleted_at	DATETIME		Yes	Soft delete timestamp
organizations				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
uuid	CHAR(36)	UNIQUE	No	Public-safe unique identifier
name	VARCHAR(255)		No	Organization name
slug	VARCHAR(150)	UNIQUE	No	Unique organization slug
primary_email	VARCHAR(255)		Yes	Billing/contact email
pricing_tier_id	BIGINT UNSIGNED	FK -> pricing_tiers.id	No	Assigned pricing tier
status	ENUM('active','trial','grace','suspended','cancelled')		No	Organization lifecycle status
current_period_start	DATE		Yes	Current billing period start

current_period_end	DATE		Yes	Current billing period end
grace_period_ends_at	DATETIME		Yes	Read-only grace period end
created_by_user_id	BIGINT UNSIGNED	FK -> users.id	No	User who created the organization
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
deleted_at	DATETIME		Yes	Soft delete timestamp
organization_roles				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
name	VARCHAR(100)		No	Role name like Owner, Manager, Developer
code	VARCHAR(50)	UNIQUE	No	System role code
description	TEXT		Yes	Role description
is_system_role	TINYINT(1)	DEFAULT 1	No	Marks predefined role
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
organization_permissions				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
module_name	VARCHAR(100)		No	Module name such as users, reports, alerts

permission_key	VARCHAR(150)	UNIQUE	No	Permission key like usage.view_team
description	TEXT		Yes	Permission meaning
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
organization_role_permissions				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
organization_role_id	BIGINT UNSIGNED	FK -> organization_roles.id	No	Assigned organization role
organization_permission_id	BIGINT UNSIGNED	FK -> organization_permissions.id	No	Assigned permission
created_at	DATETIME		No	Record creation timestamp
organization_users				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Organization membership scope
user_id	BIGINT UNSIGNED	FK -> users.id	No	Member user
organization_role_id	BIGINT UNSIGNED	FK -> organization_roles.id	No	Assigned role within organization
invitation_status	ENUM('invited','accepted','expired','revoked')		No	Invitation state
invited_by_user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Who invited the member

invited_at	DATETIME		Yes	Invitation timestamp
accepted_at	DATETIME		Yes	Acceptance timestamp
status	ENUM('active','inactive','removed')		No	Membership status
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
deleted_at	DATETIME		Yes	Soft delete timestamp
organization_invites				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Organization being joined
email	VARCHAR(255)		No	Invited email address
organization_role_id	BIGINT UNSIGNED	FK -> organization_roles.id	No	Intended role
invite_token	VARCHAR(255)	UNIQUE	No	Secure invitation token
invited_by_user_id	BIGINT UNSIGNED	FK -> users.id	No	User who sent the invite
expires_at	DATETIME		No	Invitation expiry
accepted_at	DATETIME		Yes	Acceptance timestamp
status	ENUM('pending','accepted','expired','revoked')		No	Invite state
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
api_keys				

Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
uuid	CHAR(36)	UNIQUE	No	Public-safe unique identifier
organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Owning organization
user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Nullable; NULL means organization-level key
key_name	VARCHAR(150)		No	Friendly API key name
key_prefix	VARCHAR(20)		No	Safe display prefix
key_hash	VARCHAR(255)		No	Stored hashed API key
scope_type	ENUM('organization','user')		No	Key scope
status	ENUM('active','revoked','expired')		No	API key status
last_used_at	DATETIME		Yes	Last time key was used
expires_at	DATETIME		Yes	Optional expiry
revoked_at	DATETIME		Yes	Revocation timestamp
revoked_by_user_id	BIGINT UNSIGNED	FK -> users.id	Yes	User who revoked the key
created_by_user_id	BIGINT UNSIGNED	FK -> users.id	No	User who created the key
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
api_key_to_access				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key

api_key_id	BIGINT UNSIGNED	FK -> api_keys.id	No	API key
third_party_tool_id	BIGINT UNSIGNED	FK -> third_party_tools.id	No	Allowed third-party tool
created_at	DATETIME		No	Record creation timestamp
id	BIGINT UNSIGNED	PK	No	Primary key
api_key_id	BIGINT UNSIGNED	FK -> api_keys.id	No	API key
action_type	ENUM('created','rotated','revoked','used')		No	API key event
performed_by_user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Actor user where applicable
metadata_json	JSON		Yes	Extra details
created_at	DATETIME		No	Log creation timestamp
usage_events				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
uuid	CHAR(36)	UNIQUE	No	Public-safe unique identifier
organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Organization owning the event
user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Nullable when only org-level key context exists
api_key_id	BIGINT UNSIGNED	FK -> api_keys.id	No	API key used for ingestion
third_party_tool_id	BIGINT UNSIGNED	FK -> third_party_tools.id	No	Tool that generated the event
external_event_id	VARCHAR(255)		Yes	Source event id for deduplication

source_session_id	VARCHAR(255)		Yes	Source session/conversation id
model_name	VARCHAR(150)		Yes	Model used e.g. gpt-4.1 or claude-sonnet
event_type	ENUM('completion','chat','token_usage','cost_usage','other')		No	Telemetry event classification
input_tokens	BIGINT	DEFAULT 0	No	Input token count
output_tokens	BIGINT	DEFAULT 0	No	Output token count
total_tokens	BIGINT	DEFAULT 0	No	Total token count
estimated_cost	DECIMAL(14,6)	DEFAULT 0.000000	No	Estimated cost for the event
currency_code	CHAR(3)	DEFAULT 'USD'	No	Cost currency
event_occurred_at	DATETIME		No	Actual event timestamp
received_at	DATETIME		No	Platform receive timestamp
raw_payload_json	JSON		Yes	Original telemetry payload
created_at	DATETIME		No	Record creation timestamp
usage_daily_summaries				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
summary_date	DATE		No	Summary date
organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Organization scope
user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Nullable for org-level summary

third_party_tool_id	BIGINT UNSIGNED	FK -> third_party_tools.id	Yes	Optional tool grouping
model_name	VARCHAR(150)		Yes	Optional model grouping
total_events	BIGINT	DEFAULT 0	No	Event count
total_input_tokens	BIGINT	DEFAULT 0	No	Input token total
total_output_tokens	BIGINT	DEFAULT 0	No	Output token total
total_tokens	BIGINT	DEFAULT 0	No	Total token count
total_cost	DECIMAL(14,6)	DEFAULT 0.000000	No	Total estimated cost
currency_code	CHAR(3)	DEFAULT 'USD'	No	Cost currency
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
usage_monthly_summaries				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
summary_month	DATE		No	First day of month convention
organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Organization scope
user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Nullable for org-level summary
third_party_tool_id	BIGINT UNSIGNED	FK -> third_party_tools.id	Yes	Optional tool grouping
model_name	VARCHAR(150)		Yes	Optional model grouping

total_event_s	BIGINT	DEFAULT 0	No	Event count
total_input_tokens	BIGINT	DEFAULT 0	No	Input token total
total_output_tokens	BIGINT	DEFAULT 0	No	Output token total
total_tokens	BIGINT	DEFAULT 0	No	Total token count
total_cost	DECIMAL(14,6)	DEFAULT 0.000000	No	Total estimated cost
currency_code	CHAR(3)	DEFAULT 'USD'	No	Cost currency
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
alert_rules				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
uuid	CHAR(36)	UNIQUE	No	Public-safe unique identifier
organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Organization scope
created_by_user_id	BIGINT UNSIGNED	FK -> users.id	No	User who created the alert
alert_name	VARCHAR(150)		No	User-friendly alert name
alert_type	ENUM('cost_threshold','token_threshold','tool_usage_spike','model_usage_spike')		No	Alert category
scope_type	ENUM('organization','user','tool','model')		No	Monitoring scope
scope_user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Used when scope_type = user
scope_tool_id	BIGINT UNSIGNED	FK -> third_party_tools.id	Yes	Used when scope_type = tool

scope_model_name	VARCHAR(150)		Yes	Used when scope_type = model
threshold_operator	ENUM('gte','lte','eq','gt','lt')		No	Comparison operator
threshold_value	DECIMAL(14,6)		No	Threshold numeric value
evaluation_window	ENUM('daily','weekly','monthly')		No	Evaluation window
notification_channel	ENUM('email','in_app','both')		No	Delivery channel
status	ENUM('active','inactive')		No	Alert status
last_evaluated_at	DATETIME		Yes	Last evaluation timestamp
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp
deleted_at	DATETIME		Yes	Soft delete timestamp
alert_rule_recipients				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
alert_rule_id	BIGINT UNSIGNED	FK -> alert_rules.id	No	Alert rule
user_id	BIGINT UNSIGNED	FK -> users.id	No	User to notify
created_at	DATETIME		No	Record creation timestamp
alert_trigger_logs				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
alert_rule_id	BIGINT UNSIGNED	FK -> alert_rules.id	No	Alert rule

organization_id	BIGINT UNSIGNED	FK -> organizations.id	No	Organization scope
triggered_for_user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Target user if applicable
trigger_value	DECIMAL(14,6)		No	Measured value at trigger time
threshold_value	DECIMAL(14,6)		No	Configured threshold
status	ENUM('triggered','sent','failed','ignored')		No	Trigger processing status
message_snapshot	TEXT		Yes	Rendered message at trigger time
triggered_at	DATETIME		No	Trigger timestamp
created_at	DATETIME		No	Log creation timestamp
alert_notifications				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
alert_trigger_log_id	BIGINT UNSIGNED	FK -> alert_trigger_logs.id	No	Related trigger log
recipient_user_id	BIGINT UNSIGNED	FK -> users.id	No	Recipient user
channel	ENUM('email','in_app')		No	Delivery channel
delivery_status	ENUM('pending','sent','failed','bounced','read')		No	Delivery result
sent_at	DATETIME		Yes	Sent timestamp
read_at	DATETIME		Yes	Read timestamp for in-app
failure_reason	TEXT		Yes	Failure or bounce detail
created_at	DATETIME		No	Record creation timestamp
updated_at	DATETIME		No	Record update timestamp

user_sessions				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
user_id	BIGINT UNSIGNED	FK -> users.id	No	Related user
access_token_jti	VARCHAR(255)	UNIQUE	No	JWT token identifier
refresh_token_hash	VARCHAR(255)		Yes	Hashed refresh token
ip_address	VARCHAR(45)		Yes	IPv4/IPv6 address
user_agent	TEXT		Yes	Browser/device details
expires_at	DATETIME		No	Session expiry
revoked_at	DATETIME		Yes	Revocation timestamp
created_at	DATETIME		No	Record creation timestamp
user_audit_logs				
Field Name	Type	Key Constraint	Nullable	Description
id	BIGINT UNSIGNED	PK	No	Primary key
organization_id	BIGINT UNSIGNED	FK -> organizations.id	Yes	Organization context
user_id	BIGINT UNSIGNED	FK -> users.id	Yes	Actor user
action_type	VARCHAR(100)		No	Action such as CREATE, UPDATE, LOGIN
module_name	VARCHAR(100)		No	Affected module
record_type	VARCHAR(100)		No	Target entity/table name
record_id	VARCHAR(100)		No	Target record id

old_values_ json	JSON		Yes	Previous state snapshot
new_values_ json	JSON		Yes	Updated state snapshot
ip_address	VARCHAR(45)		Yes	IPv4/IPv6 address
user_agent	TEXT		Yes	Request user agent
created_at	DATETIME		No	Log creation timestamp

API Structure Proposal

Below is the Proposed API Structure Document for V1:

- **Repository 1:** Backend API only
- **Repository 2:** Admin Setup frontend only
- **Repository 3:** User Setup frontend only
- Same backend base URL
- Separate auth endpoints and JWT scopes
- JWT access token + refresh token
- **Separate identities** for admin and organization users
- **REST API** only
- Backend validation mandatory on all write APIs

This structure is shaped around the PRD flows for onboarding, integrations, usage dashboards, invites, reports, API key setup, alerting, and role-based organization management.

Admin Authentication APIs

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Admin Auth	Admin Login	POST	/api/v1/admin/auth/login	Login for platform admin users and issue access + refresh token	Repo 1 → Repo 2
Admin Auth	Admin Refresh Token	POST	/api/v1/admin/auth/refresh	Issue new access token from valid refresh token	Repo 1 → Repo 2
Admin Auth	Admin Logout	POST	/api/v1/admin/auth/logout	Revoke current refresh/session token	Repo 1 → Repo 2
Admin Auth	Admin Profile	GET	/api/v1/admin/auth/me	Fetch logged-in admin profile, roles, permissions, scope	Repo 1 → Repo 2

Admin Auth	Change Password	POST	<code>/api/v1/admin/auth/change-password</code>	Change admin password with old password validation	Repo 1 → Repo 2
Admin Auth	Forgot Password	POST	<code>/api/v1/admin/auth/forgot-password</code>	Initiate password reset for admin login	Repo 1 → Repo 2
Admin Auth	Reset Password	POST	<code>/api/v1/admin/auth/reset-password</code>	Complete admin password reset using token	Repo 1 → Repo 2

Admin User / Role / Permission APIs

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Admin Users	List Admin Users	GET	<code>/api/v1/admin/users</code>	List platform admin users with filters and pagination	Repo 1 → Repo 2
Admin Users	Create Admin User	POST	<code>/api/v1/admin/users</code>	Create new admin identity	Repo 1 → Repo 2
Admin Users	Get Admin User	GET	<code>/api/v1/admin/users/:adminUserId</code>	Fetch admin user details	Repo 1 → Repo 2
Admin Users	Update Admin User	PATCH	<code>/api/v1/admin/users/:adminUserId</code>	Update admin profile/status	Repo 1 → Repo 2
Admin Users	Delete Admin User	DELETE	<code>/api/v1/admin/users/:adminUserId</code>	Soft delete	Repo 1 → Repo 2

				admin user	
Admin Roles	List Admin Roles	GET	/api/v1/admin/roles	List admin roles	Repo 1 → Repo 2
Admin Roles	Create Admin Role	POST	/api/v1/admin/roles	Create admin role	Repo 1 → Repo 2
Admin Roles	Get Admin Role	GET	/api/v1/admin/roles/:roleId	Fetch role details with permissions	Repo 1 → Repo 2
Admin Roles	Update Admin Role	PATCH	/api/v1/admin/roles/:roleId	Update admin role metadata	Repo 1 → Repo 2
Admin Roles	Delete Admin Role	DELETE	/api/v1/admin/roles/:roleId	Soft delete role if not protected	Repo 1 → Repo 2
Admin Permissions	List Admin Permissions	GET	/api/v1/admin/permissions	List permission master records	Repo 1 → Repo 2
Admin Role Permissions	Assign Permissions to Role	PUT	/api/v1/admin/roles/:roleId/permissions	Replace/edit permission mapping for a role	Repo 1 → Repo 2
Admin User Roles	Assign Roles to Admin User	PUT	/api/v1/admin/users/:adminUserId/roles	Attach one or more admin roles to admin user	Repo 1 → Repo 2

Pricing Tier Setup APIs

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Pricing Tiers	List Pricing Tiers	GET	/api/v1/admin/pricing-tiers	List pricing tiers	Repo 1 → Repo 2
Pricing Tiers	Create Pricing Tier	POST	/api/v1/admin/pricing-tiers	Create tier like Free / Pro / Enterprise	Repo 1 → Repo 2
Pricing Tiers	Get Pricing Tier	GET	/api/v1/admin/pricing-tiers/:tierId	Fetch pricing tier details	Repo 1 → Repo 2
Pricing Tiers	Update Pricing Tier	PATCH	/api/v1/admin/pricing-tiers/:tierId	Update limits and pricing config	Repo 1 → Repo 2
Pricing Tiers	Delete Pricing Tier	DELETE	/api/v1/admin/pricing-tiers/:tierId	Soft delete/inactivate tier	Repo 1 → Repo 2
Pricing Features	List Tier Feature Overrides	GET	/api/v1/admin/pricing-tiers/:tierId/features	Fetch flexible feature config for a tier	Repo 1 → Repo 2
Pricing Features	Upsert Tier Feature Overrides	PUT	/api/v1/admin/pricing-tiers/:tierId/features	Maintain feature-value overrides	Repo 1 → Repo 2

Third-Party Tool Setup APIs

The PRD specifically depends on integrations for tools like Claude, Codex, Gemini, and similar AI coding tools.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
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Tool Master	List Tools	GET	<code>/api/v1/admin/tools</code>	List supported third-party AI tools	Repo 1 → Repo 2
Tool Master	Create Tool	POST	<code>/api/v1/admin/tools</code>	Create tool master record	Repo 1 → Repo 2
Tool Master	Get Tool	GET	<code>/api/v1/admin/tools/:toolId</code>	Fetch tool details	Repo 1 → Repo 2
Tool Master	Update Tool	PATCH	<code>/api/v1/admin/tools/:toolId</code>	Update metadata/status/integration type	Repo 1 → Repo 2
Tool Master	Delete Tool	DELETE	<code>/api/v1/admin/tools/:toolId</code>	Soft delete/inactivate tool	Repo 1 → Repo 2
Tool Config	List Tool Configs	GET	<code>/api/v1/admin/tools/:toolId/configs</code>	List platform-level tool configs	Repo 1 → Repo 2
Tool Config	Create Tool Config	POST	<code>/api/v1/admin/tools/:toolId/configs</code>	Add config key/value for tool	Repo 1 → Repo 2
Tool Config	Update Tool Config	PATCH	<code>/api/v1/admin/tools/:toolId/configs/:configId</code>	Update tool config value	Repo 1 → Repo 2
Tool Config	Delete Tool Config	DELETE	<code>/api/v1/admin/tools/:toolId/configs/:configId</code>	Remove config entry	Repo 1 → Repo 2

Platform Organization Oversight APIs

The PRD requires organization lifecycle handling, seat limits, invitation blocking on plan limits, read-only grace period on cancellation, and platform visibility into spend and usage.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
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Organizations Admin	List Organizations	GET	<code>/api/v1/admin/organizations</code>	List customer organizations with filters	Repo 1 → Repo 2
Organizations Admin	Get Organization	GET	<code>/api/v1/admin/organizations/:organizationId</code>	Fetch organization details including tier and status	Repo 1 → Repo 2
Organizations Admin	Update Organization Status	PATCH	<code>/api/v1/admin/organizations/:organizationId/status</code>	Suspend, activate, grace, cancel org	Repo 1 → Repo 2
Organizations Admin	Change Organization Tier	PATCH	<code>/api/v1/admin/organizations/:organizationId/pricing-tier</code>	Move organization between tiers	Repo 1 → Repo 2
Organizations Admin	List Organization Members	GET	<code>/api/v1/admin/organizations/:organizationId/members</code>	Platform view into org membership	Repo 1 → Repo 2
Organizations Admin	Revoke Organization API Key	POST	<code>/api/v1/admin/organizations/:organizationId/api-keys/:apiKeyId/revoke</code>	Force revoke org/user key under platform control	Repo 1 → Repo 2

Platform Usage / Alert Oversight APIs

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Platform Usage	Platform Usage Summary	GET	/api/v1/admin/usage/summary	Global usage overview across all organizations	Repo 1 → Repo 2
Platform Usage	Organization Usage Breakdown	GET	/api/v1/admin/usage/organizations/:organizationId	Detailed usage/cost breakdown for one org	Repo 1 → Repo 2
Platform Alerts	List Triggered Alerts	GET	/api/v1/admin/alerts/triggers	View alert trigger history across tenants	Repo 1 → Repo 2
Platform Reports	Export Organization Usage CSV	POST	/api/v1/admin/reports/organizations/:organizationId/export	Queue large CSV export for an organization	Repo 1 → Repo 2
Platform Reports	Export Job Status	GET	/api/v1/admin/reports/exports/:jobId	Check async export status and download link metadata	Repo 1 → Repo 2

App Authentication APIs

The PRD requires self-signup with organization creation, invite-based joining, and shared organization dashboards.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
App Auth	User Signup + Create Organization	POST	/api/v1/app/auth/signup	Create user account and first organization in one flow	Repo 1 → Repo 3
App Auth	User Login	POST	/api/v1/app/auth/login	Login organization user and issue access + refresh token	Repo 1 → Repo 3
App Auth	Refresh Token	POST	/api/v1/app/auth/refresh	Refresh app access token	Repo 1 → Repo 3
App Auth	Logout	POST	/api/v1/app/auth/logout	Revoke active refresh/session token	Repo 1 → Repo 3
App Auth	Verify Email	POST	/api/v1/app/auth/verify-email	Verify user email with token/OTP flow	Repo 1 → Repo 3
App Auth	Forgot Password	POST	/api/v1/app/auth/forgot-password	Start password reset	Repo 1 → Repo 3
App Auth	Reset Password	POST	/api/v1/app/auth/reset-password	Complete password reset	Repo 1 → Repo 3
App Auth	My Profile	GET	/api/v1/app/auth/me	Fetch authenticated user profile, org, roles, permissions	Repo 1 → Repo 3

Invitation and Organization Join APIs

The PRD explicitly includes invitation flow, invite acceptance, pending users, and invitee joining a shared organization dashboard.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Invites	Validate Invite Token	GET	<code>/api/v1/app/invites/:token</code>	Check whether invite token is valid, pending, expired, or revoked	Repo 1 → Repo 3
Invites	Accept Invite	POST	<code>/api/v1/app/invites/:token/accept</code>	Accept organization invitation and join team	Repo 1 → Repo 3
Invites	Resend Invite	POST	<code>/api/v1/app/organization/invites/:inviteId/resend</code>	Resend invite email	Repo 1 → Repo 3
Invites	Revoke Invite	POST	<code>/api/v1/app/organization/invites/:inviteId/revoke</code>	Revoke pending invite	Repo 1 → Repo 3

Organization Profile / Membership / RBAC APIs

The PRD calls for owner-driven team management, role assignment, role changes, removal, and prevention of orphaned ownership.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Organization	Get Organi	GET	<code>/api/v1/app/organization</code>	Fetch current organization details	Repo 1 →

	zation Profile				Repo 3
Organi zation	Update Organi zation Profile	PAT CH	/api/v1/app/organization	Update organization metadata allowed in app scope	Repo 1 → Repo 3
Organi zation Membe rs	List Membe rs	GET	/api/v1/app/organization/members	List users in current organization	Repo 1 → Repo 3
Organi zation Membe rs	Invite Membe r	POS T	/api/v1/app/organization/members/invite	Invite user to organization with selected role	Repo 1 → Repo 3
Organi zation Membe rs	Get Membe r	GET	/api/v1/app/organization/members/:memberId	Fetch one member	Repo 1 → Repo 3
Organi zation Membe rs	Chang e Membe r Role	PAT CH	/api/v1/app/organization/members/:memberId/role	Change Owner/Manager/ Developer role mapping	Repo 1 → Repo 3
Organi zation Membe rs	Remov e Membe r	DEL ETE	/api/v1/app/organization/members/:memberId	Remove member from organization	Repo 1 → Repo 3
Organi zation Roles	List Roles	GET	/api/v1/app/organization/roles	List default org roles and assigned permissions	Repo 1 → Repo 3
Organi zation Roles	Get Role Permis sions	GET	/api/v1/app/organization/roles/:roleId/permissions	View permissions assigned to a role	Repo 1 → Repo 3
Organi zation Roles	Update Role Permis sions	PUT	/api/v1/app/organization/roles/:roleId/permissions	Update editable permission mapping for default roles	Repo 1 → Repo 3

Organization Permissions	List Permission Master	GET	<code>/api/v1/app/organization/permissions</code>	List permission keys available for org-side mapping	Repo 1 → Repo 3
Organization Account	Leave Organization	POST	<code>/api/v1/app/organization/leave</code>	User leaves current organization if allowed by business rules	Repo 1 → Repo 3

Critical backend sanity checks

- prevent last remaining Owner from leaving or being removed
- prevent role downgrade that leaves org with zero Owners
- reject invite when seat limit is exceeded for plan
- reject duplicate active membership
- if invited user already belongs to another org, enforce leave/remove-first flow

API Key Management APIs

The PRD requires API key generation, copy/paste into tool config, and instant revocation/replacement without data loss.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
API Keys	List API Keys	GET	<code>/api/v1/app/api-keys</code>	List org-level and user-level API keys for current organization	Repo 1 → Repo 3
API Keys	Create API Key	POST	<code>/api/v1/app/api-keys</code>	Create new API key for organization or user scope	Repo 1 → Repo 3
API Keys	Get API Key	GET	<code>/api/v1/app/api-keys/:apiKeyId</code>	Fetch safe metadata for one API key	Repo 1 → Repo 3
API Keys	Revoke API Key	POST	<code>/api/v1/app/api-keys/:apiKeyId/revoke</code>	Revoke API key immediately	Repo 1 → Repo 3

API Keys	Rotate API Key	POST	<code>/api/v1/app/api-keys/:apiKeyId/rotate</code>	Revoke old key and issue replacement	Repo 1 → Repo 3
API Keys	Delete API Key	DELETE	<code>/api/v1/app/api-keys/:apiKeyId</code>	Soft delete inactive/revoked key metadata if policy allows	Repo 1 → Repo 3
API Keys	List Allowed Tools for Key	GET	<code>/api/v1/app/api-keys/:apiKeyId/tools</code>	View tool linkage for a key	Repo 1 → Repo 3
API Keys	Assign Tool Access to Key	PUT	<code>/api/v1/app/api-keys/:apiKeyId/tools</code>	Maintain allowed tool mapping if used in schema	Repo 1 → Repo 3

Tool Discovery APIs for App Frontend

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Integrations	List Supported Tools	GET	<code>/api/v1/app/tools</code>	List supported tools for user integration setup	Repo 1 → Repo 3
Integrations	Get Tool Setup Guide	GET	<code>/api/v1/app/tools/:toolId/setup</code>	Return frontend-friendly integration instructions and metadata	Repo 1 → Repo 3

Telemetry Ingestion APIs

The PRD assumes AI tools can send telemetry using API key integration and that first telemetry should populate dashboard usage.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Ingestion	Ingest Usage Event	POST	<code>/api/v1/ingestion/events</code>	Accept one telemetry usage event from server-to-server integration	Repo 1
Ingestion	Bulk Ingest Usage Events	POST	<code>/api/v1/ingestion/events/bulk</code>	Accept multiple events in one request	Repo 1
Ingestion	Validate API Key Connectivity	POST	<code>/api/v1/ingestion/validate-key</code>	Lightweight validation endpoint for integration testing	Repo 1
Ingestion	Health Check	GET	<code>/api/v1/ingestion/health</code>	Health endpoint for integration monitoring	Repo 1

Mandatory backend validations

- API key exists and is active
- tool is supported
- org is active/grace policy allows ingestion decision
- event timestamp is valid
- token counts and cost values are non-negative
- reject impossible totals
- dedupe using `external_event_id` where supplied
- normalize tool/model names
- store raw payload and derived normalized cost fields

Dashboard / Usage Analytics APIs

The PRD requires dashboard summaries, daily cost, token usage, model breakdown, filtering by date/tool/model, and team visibility.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Dashboard	Summary Cards	GET	/api/v1/app/dashboard/summary	Return high-level metrics like cost, tokens, top tool, top model	Repo 1 → Repo 3
Dashboard	Usage Trend	GET	/api/v1/app/dashboard/usage-trend	Return time-series usage/cost for selected date range	Repo 1 → Repo 3
Dashboard	Tool Breakdown	GET	/api/v1/app/dashboard/tools	Cost and tokens grouped by tool	Repo 1 → Repo 3
Dashboard	Model Breakdown	GET	/api/v1/app/dashboard/models	Cost and tokens grouped by model	Repo 1 → Repo 3
Dashboard	User Breakdown	GET	/api/v1/app/dashboard/users	Team usage by member, permission-dependent	Repo 1 → Repo 3
Dashboard	Recent Events	GET	/api/v1/app/dashboard/events	Recent normalized usage events with filters	Repo 1 → Repo 3

Common query filters

- [from_date](#)
- [to_date](#)
- [tool_id](#) or [tool_code](#)
- [model_name](#)
- [user_id](#)
- [group_by](#)

Reports and CSV Export APIs

The PRD explicitly requires monthly reports and CSV export, and also flags CSV size constraints as an edge case.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Reports	Monthly Report Summary	GET	<code>/api/v1/app/reports/monthly-summary</code>	Return report-ready summary for selected month/date range	Repo 1 → Repo 3
Reports	Weekly Digest Summary	GET	<code>/api/v1/app/reports/weekly-digest</code>	Return weekly digest style metrics	Repo 1 → Repo 3
Reports	Queue CSV Export	POST	<code>/api/v1/app/reports/exports/csv</code>	Queue CSV generation for heavy usage exports	Repo 1 → Repo 3
Reports	Export Job Status	GET	<code>/api/v1/app/reports/exports/:jobId</code>	Check queued export status	Repo 1 → Repo 3
Reports	Download Export Metadata	GET	<code>/api/v1/app/reports/exports/:jobId/download</code>	Provide file metadata/download token when job is completed	Repo 1 → Repo 3

Alerts APIs

The PRD requires spend threshold alerting before invoice arrival, user-created thresholds, and handling alert delivery failure such as email bounce.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Alerts	List Alert Rules	GET	/api/v1/app/alerts	List active/inactive alert rules in current organization	Repo 1 → Repo 3
Alerts	Create Alert Rule	POST	/api/v1/app/alerts	Create organization-wide or personal threshold alert	Repo 1 → Repo 3
Alerts	Get Alert Rule	GET	/api/v1/app/alerts/:alertId	Fetch one alert rule	Repo 1 → Repo 3
Alerts	Update Alert Rule	PATCH	/api/v1/app/alerts/:alertId	Update threshold/scope/channel/status	Repo 1 → Repo 3
Alerts	Delete Alert Rule	DELETE	/api/v1/app/alerts/:alertId	Soft delete alert rule	Repo 1 → Repo 3
Alerts	List Alert Triggers	GET	/api/v1/app/alerts/triggers	View alert trigger history	Repo 1 → Repo 3
Alerts	Get Alert Trigger	GET	/api/v1/app/alerts/triggers/:triggerId	Fetch one trigger event with delivery details	Repo 1 → Repo 3
Alerts	Retry Failed Alert Email	POST	/api/v1/app/alerts/triggers/:triggerId/retry	Retry email for failed delivery if policy allows	Repo 1 → Repo 3

Profile / Session APIs

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Profile	Get My Profile	GET	/api/v1/app/profile	Return user profile and org membership info	Repo 1 → Repo 3
Profile	Update My Profile	PATCH	/api/v1/app/profile	Update personal profile fields	Repo 1 → Repo 3
Sessions	List Active Sessions	GET	/api/v1/app/sessions	List active login sessions	Repo 1 → Repo 3
Sessions	Revoke Session	DELETE	/api/v1/app/sessions/:sessionId	Revoke selected session	Repo 1 → Repo 3

Background Job APIs / Internal Schedulers

These may be internal-only, not frontend-consumed.

Module	Endpoint Name	Method	URL	Description	Repository Mapping
Internal Jobs	Run Daily Usage Aggregation	POST	/api/v1/internal/jobs/aggregate/daily	Aggregate raw events into daily summaries	Repo 1
Internal Jobs	Run Monthly Usage Aggregation	POST	/api/v1/internal/jobs/aggregate/monthly	Aggregate monthly summaries	Repo 1

Internal Jobs	Evaluate Alert Rules	POST	<code>/api/v1/internal/jobs/alerts/evaluate</code>	Evaluate scheduled threshold alerts	Repo 1
Internal Jobs	Process Export Queue	POST	<code>/api/v1/internal/jobs/reports/process-exports</code>	Generate queued CSV exports	Repo 1
Internal Jobs	Cleanup Expired Invites / Tokens	POST	<code>/api/v1/internal/jobs/cleanup/auth</code>	Cleanup expired auth and invite records	Repo 1

Recommended Permission Ownership

Admin Scope

Use admin RBAC for:

- admin user CRUD
- admin role/permission mapping
- pricing tiers
- tool master/config
- platform organization oversight
- global usage and export access
- force-revoke org keys
- org suspend/activate actions

Organization Scope

Use org RBAC for:

- member invites
- role changes
- API key creation/revocation
- dashboard views
- report exports
- alert creation and review

Suggested initial organization permissions

- organization.view
 - organization.update
 - members.view
 - members.invite
 - members.role.update
 - members.remove
 - roles.view
 - roles.permissions.update
 - api_keys.view
 - api_keys.create
 - api_keys.revoke
 - usage.view.self
 - usage.view.team
 - reports.view
 - reports.export
 - alerts.view
 - alerts.create
 - alerts.update
 - alerts.delete
-

Recommended Auth / Scope Behavior

Admin JWT

Claims:

- sub
- scope=platform_admin
- admin_user_id
- roles
- permissions

App JWT

Claims:

- sub
- scope=organization_user
- user_id
- organization_id
- organization_role
- permissions

Refresh Token

Store hashed refresh token or token family record in backend DB.

Backend Sanity Checks That Must Exist

Auth

- block inactive/locked admin accounts
- block inactive/suspended users
- reject refresh token replay
- support password reset expiry

Organization Membership

- one active organization membership at a time
- pending invites cannot create duplicate memberships
- last Owner protection
- seat cap check before invite acceptance or invite creation, depending on chosen policy

API Keys

- never return full key again after creation screen
- revoke immediately
- rotation preserves historical usage
- org-level key and user-level key both supported

Usage Ingestion

- validate tool ownership/support
- dedupe repeated events
- normalize costs
- store raw event plus normalized fields
- queue heavy post-processing, do not block ingestion unnecessarily

Alerts

- alert scope must match creator permission
- email-only delivery in V1
- failed delivery must be tracked
- scheduled evaluation window must be enforced

Third-Party Dependencies

The following dependencies are suggested. Any addition outside this list requires an architecture review.

Service / Package	Purpose		Alternatives
PostgreSQL 16	Primary data store	Proven reliability	MySQL
Redis 7	Cache, session store, job queue, rate limiter	Standard pairing with Node.js; single dependency covers four concerns	
Resend (or AWS SES)	Transactional email	Simple API; high deliverability; bounce webhook support	
BullMQ	Background job processing	Redis-backed; retry / backoff / delay; active maintenance	
Stripe	Subscription billing	Industry standard; hosted checkout reduces PCI scope	
Cloudflare	CDN, DDoS, TLS	Free tier covers Phase 1 needs entirely	AWS CloudFront